



SOCIAL SCIENCES

Ledger-Anchored Peer Review: Auditable Editorial Decisions at Scale

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ABSTRACT

Editorial decisions in scholarly publishing are largely invisible to the people they affect. We describe a protocol that anchors every editorial transition in an append-only record and prints a cryptographic digest of that record on the published article.

Across fourteen journals and 41,882 transitions, making the record legible to authors reduced median unexplained latency by 31%.

Keywords peer review · open science · auditability · editorial workflow

01 Introduction

Scientific publishing has accumulated a set of practices whose costs are borne unevenly across the system. The reproducibility of an editorial decision is rarely auditable after the fact, and readers are asked to trust a process they cannot inspect [1]. Where a manuscript went, who saw it, how long each stage took and on what grounds a decision was reached remain private to the publisher.

This paper describes a ledger-anchored review protocol in which every editorial transition is written to an append-only record, and each published article carries a cryptographic digest of that record.

Scope and terminology

Throughout, *transition* means any recorded change of manuscript state, and **digest** means the SHA-256 fingerprint of the canonical serialisation of the record.

02 Methods

We instrumented fourteen journals across three publishers over thirty months, capturing 41,882 state transitions. Instrumentation was passive: the protocol observes the editorial system rather than replacing it

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Peer review

Double-anon
referee rep
sciledger.org



Figure 1. Distribution of review latency across the instrumented cohort, by discipline and review model.

Table 1. Summary statistics for the instrumented cohort.

Metric	Median	P90
Days to first decision	42	121
Referee reports per manuscript	2	4
Rounds to acceptance	2	3

03 Main Discussion

The results indicate a systematic reduction in unexplained latency once the record is visible to authors [1], consistent with earlier cohort work [2–4]. Two mechanisms plausibly account for it, and they are not mutually exclusive.

- Visibility changes the cost of inaction for editors.
- Authors escalate earlier when a stall is legible.

Limitations

Sample bias remains: the instrumented journals opted in, and opting in may correlate with editorial hygiene.

04 Conclusion

An append-only editorial record is cheap to maintain and changes the incentives of everyone touching the manuscript.

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AUTHOR CONTRIBUTIONS

Leonardo Ferreira Cabral: Conceptualization, Methodology, Writing - original draft.
Marina Alves Rocha: Formal analysis, Data curation. **Tiago Nunes:** Software, Validation.

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DATA AVAILABILITY

Analysis code and event stream: <https://doi.org/10.5281/zenodo.0000000>

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

REFERENCES

1. Smith, J. and Oyelaran, B. (2021). *Trust without inspection*. *Nature Human Behaviour*, 5(4), 431–440. <https://doi.org/10.1038/s41562-021-01057-0>. ↩1 ↩2
2. Doe, A. (2022). Latency in peer review: a cohort study. *Science*, 376(6590), 220–230. [doi:10.1126/science.abc1234](https://doi.org/10.1126/science.abc1234) ↩1 ↩2
3. Vasconcelos, H. (2020). Append-only records for editorial workflows. *PLOS ONE*, 15(3), e0229123. [10.1371/journal.pone.0229123](https://doi.org/10.1371/journal.pone.0229123) ↩1
4. H. Vasconcelos, K. Imai (2023). *Registered reports and the credibility revolution*. *SciLedger*. <https://doi.org/10.1234/sciledger.2023.44> ↩1

VERIFIED EDITORIAL RECORD

digest 069d558396ac48d3



Verify at sciledger.org/verify/paper-uuid-0001
Review rounds **2 · 3 referee reports**
Decision **Accepted 11 Jul 2026**
Events recorded **6 transitions · append-only**

Scan to inspect the complete, tamper-evident editorial history of this article.